Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #36 | Retail, E-Commerce & Finance Restaurant Table Booking & Pre-Ordering App

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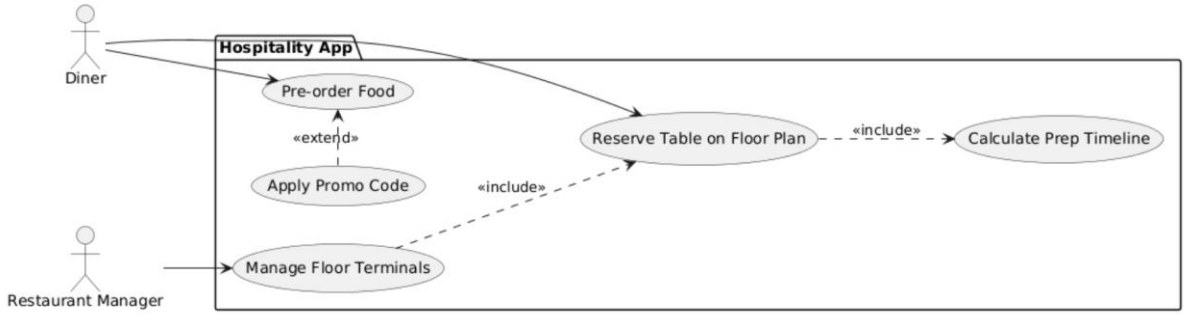
REQUIREMENTS TABLE-

ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall allow diners to pick an open table from an interactive 2D floor plan and link pre-ordered menu items to the reservation.	High	**Pass:** Table reserved and kitchen pre-order ticket queued.	Core application function.
FR-002	Functional	The system shall calculate kitchen preparation timelines to align with the diner's scheduled arrival time.	High	**Pass:** Kitchen receives automated prep alerts strictly based on diner transit times.	Ensures seamless arrival dining.
FR-003	Functional	Diners shall be able to view, modify, or cancel their active reservations and pre-orders.	Medium	**Pass:** Modifications update the kitchen queue and floor plan immediately.	Provides necessary user flexibility.

FR-004	Functional	The Restaurant Manager shall monitor live table statuses and incoming pre-orders on a centralized dashboard.	High	**Pass:** Dashboard reflects real-time reservations without manual page refreshing	Required for operational control.
FR-005	Functional	The system shall generate a unified digital invoice combining the pre-ordered food and table booking elements.	High	**Pass:** Invoice correctly sums both data points at checkout.	Streamlines the final payment process.
NR-001	Perf. & security	Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms.	High	**Pass:** Benchmarking tests confirm target latency and security standards under simulated peak load.	Prevents staff confusion and double-booking.
NR-002	Usability	The interactive 2D floor plan must be fully responsive and load within 2 seconds on	Medium	**Pass:** Load times verified under 2s on standard 4G networks.	Reduces application drop-off rates.

		mobile data.			

USE CASE DIAGRAM-



USE CASE FLOW-

Preconditions

- Diner is authenticated and logged into the application.
- The restaurant's live floor plan is currently active and accepting bookings.
- The selected table is marked as open and not already locked by another diner.

Postconditions (on success)

- Table status is updated from open to Reserved on all manager floor terminals.
- Pre-ordered menu items are permanently linked to the reservation.
- Kitchen pre-order ticket is queued with an aligned preparation timeline.

Main Success Scenario

1. Diner selects an open table from the interactive 2D floor plan.
2. System temporarily locks the table to prevent booking conflicts with other users.
3. Diner adds food items to their pre-order cart and confirms their estimated arrival time.
4. System «include» invokes Calculate Prep Timeline (UC-03) to schedule the kitchen ticket based on the arrival time.

5. Diner confirms the reservation and pre-order details.
6. System updates the live floor plan to show the table as Reserved and notifies the Restaurant Manager.
7. System queues the pre-order ticket in the kitchen system.
8. Diner receives a booking confirmation with a reservation ID.
9. Use case ends successfully.

Alternate Flow – 5a. Table Lock Expiration

1. The diner exceeds the 10-minute idle timeout during the pre-ordering process.
2. System automatically expires the temporary table lock.
3. System reverts the table status to open on the live floor plan.
4. Diner is prompted with a timeout message and asked to re-select a table to continue. Use case ends in failure.